Flipkart Price Fluctuation — EDA Report

1. Dataset Overview

What does this dataset actually cover?

- Rows: 3463
- Products: 40 across 3 categories, 7 subcategories
- Date range: 2026-08-22 00:00:00 to 2026-09-23 00:00:00

	sp	mrp	discount_percent	rating
count	3463.00	3017.00	3017.00	3463.00
mean	42550.98	55444.57	29.62	4.37
std	49654.89	63967.22	19.39	0.22
min	158.00	195.00	1.00	3.70
25%	3200.00	4999.00	14.00	4.20
50%	22999.00	29999.00	30.00	4.40
75%	66140.00	76990.00	39.00	4.50
max	215990.00	287990.00	83.00	4.80

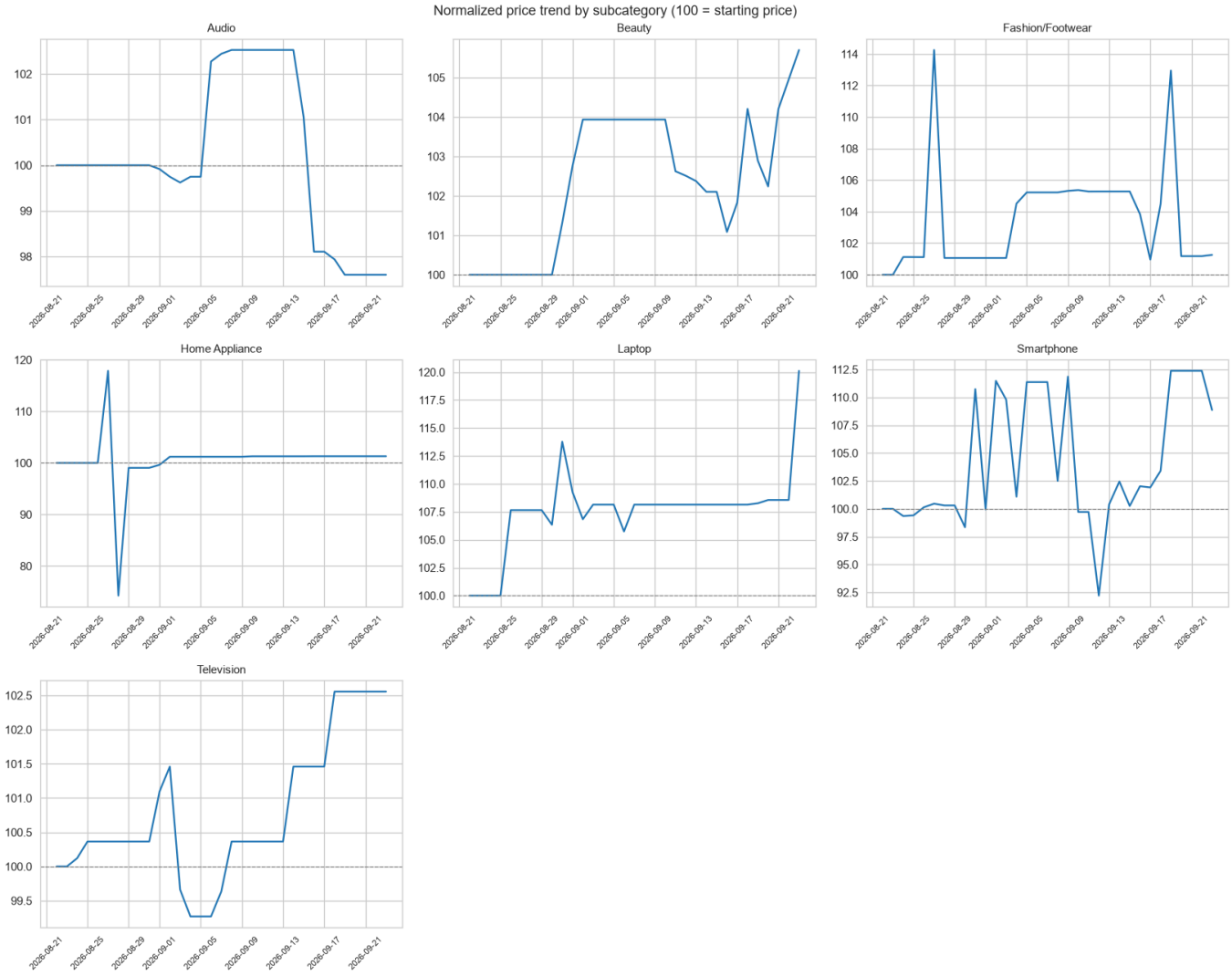
Missing values

	missing_count	missing_pct
mrp	446	12.9
discount_percent	446	12.9
bank_offer_price	53	1.5
festival_name	1408	40.7
price_change	40	1.2
price_change_pct	40	1.2
rolling_price_mean	41	1.2
rolling_price_std	82	2.4

	missing_count	missing_pct
computed_discount_percent	446	12.9
discount_discrepancy	446	12.9

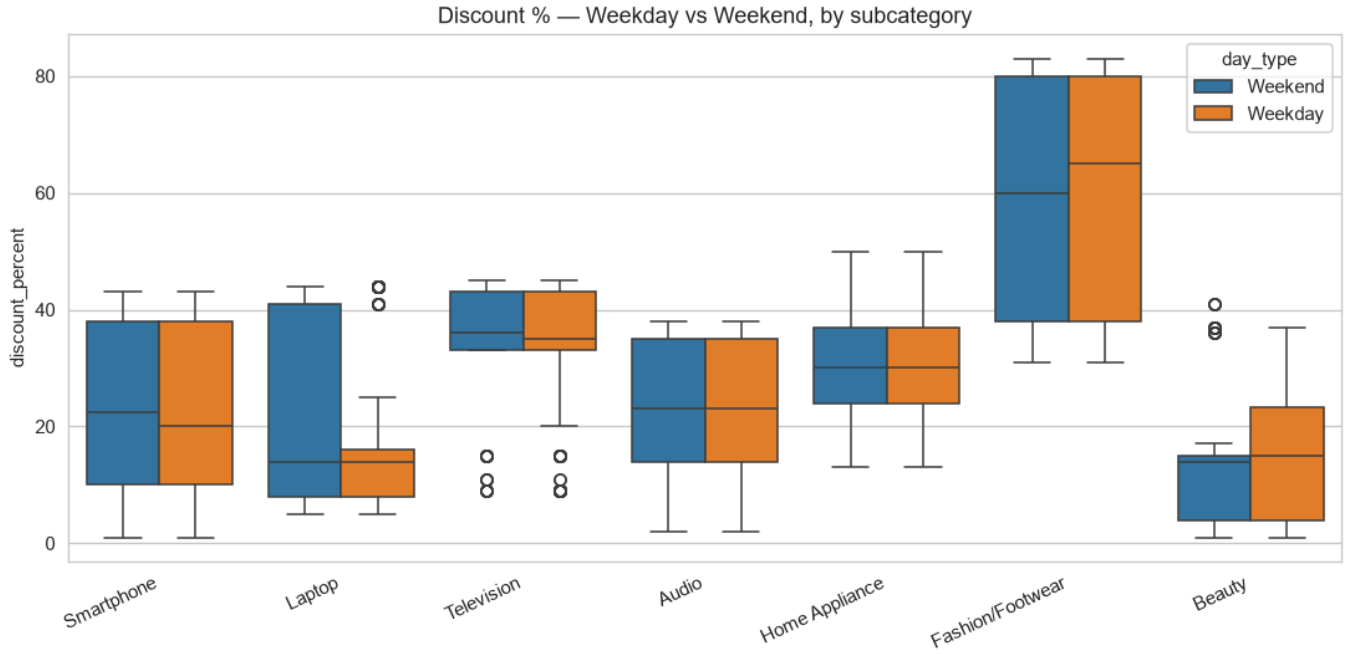
2. Price Trend

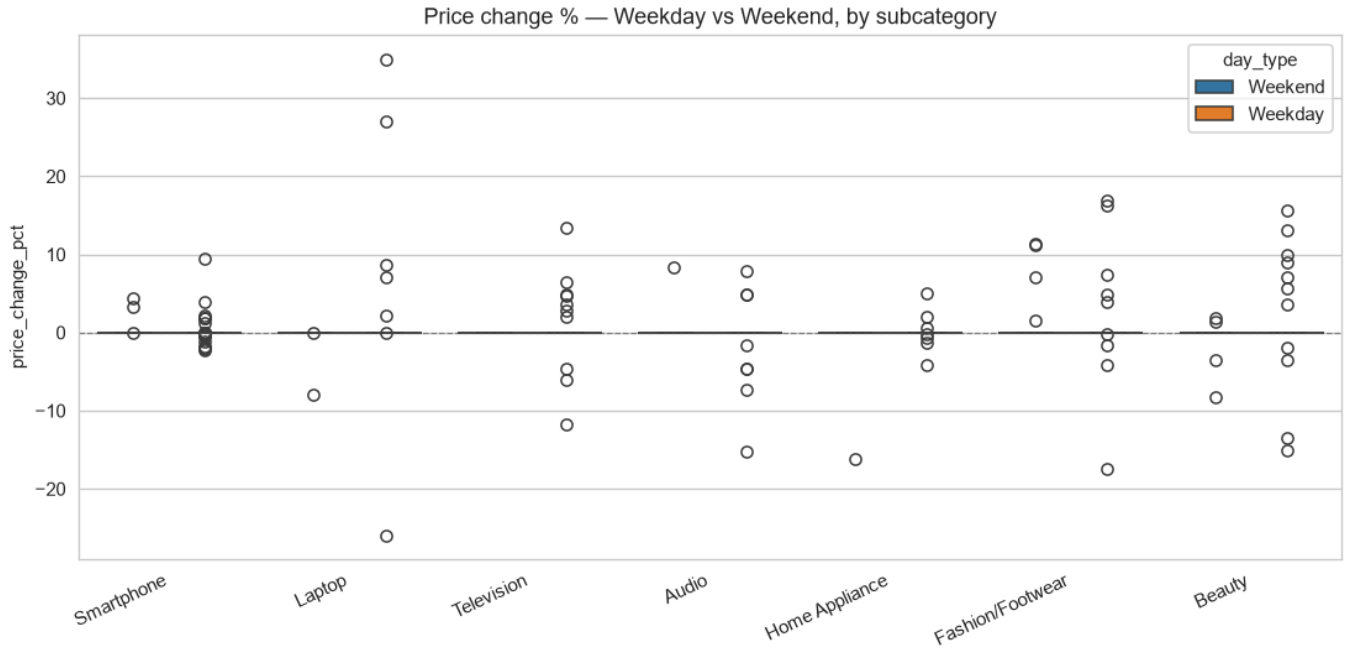
How does price move over time, WITHIN each subcategory (normalized so different price scales are comparable)?



3. Weekday vs Weekend

Within each subcategory, does discount or price movement differ weekday vs weekend?

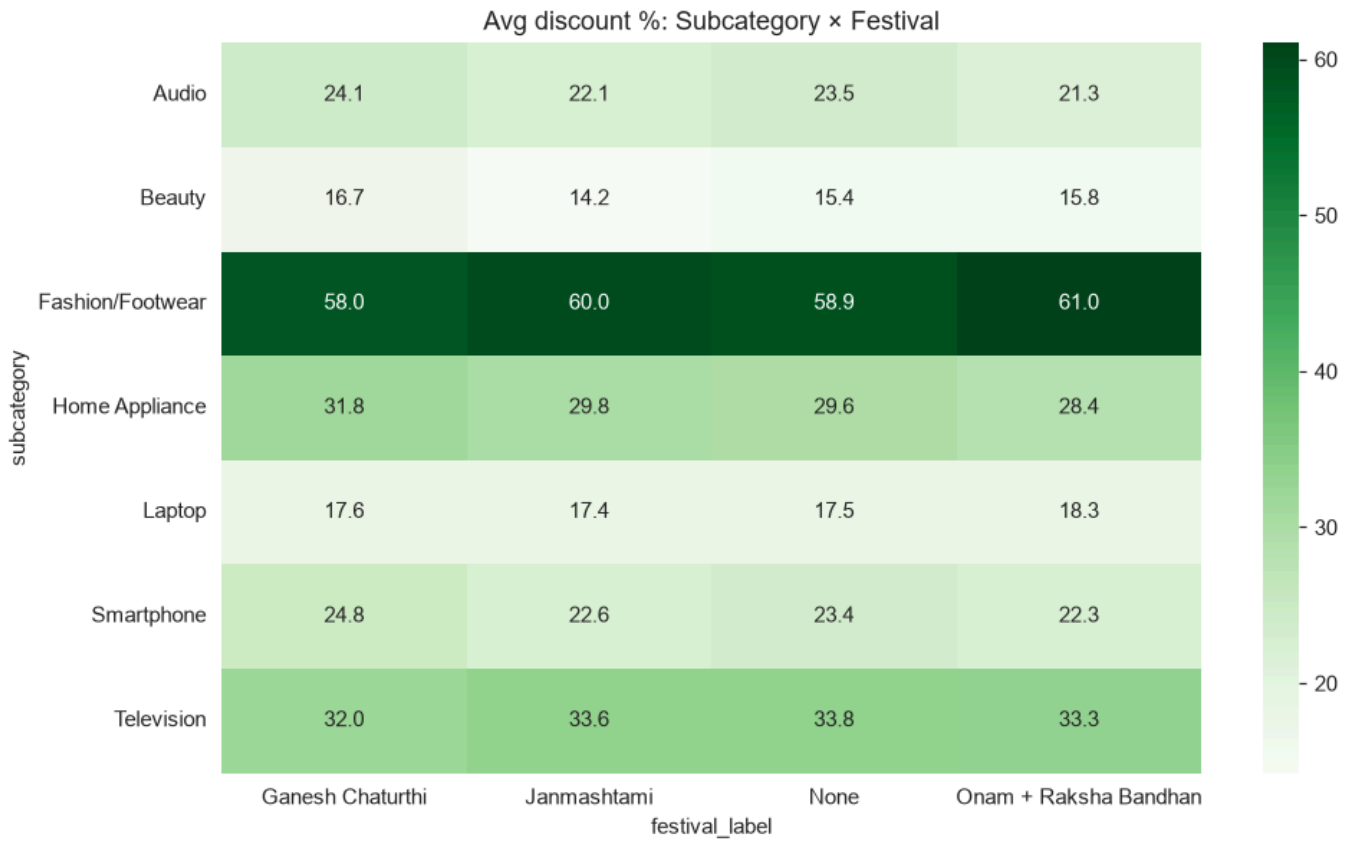


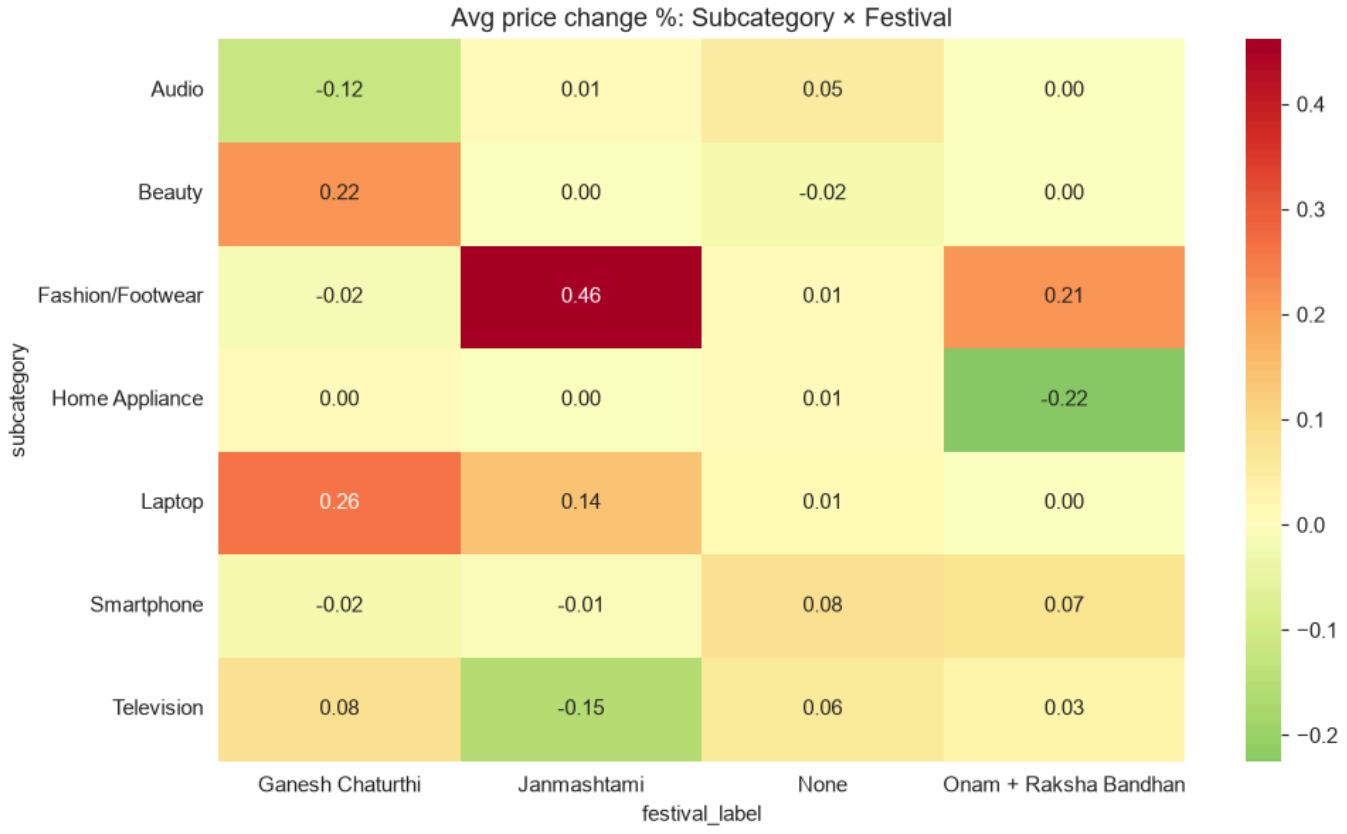


---

4. Festival Behaviour

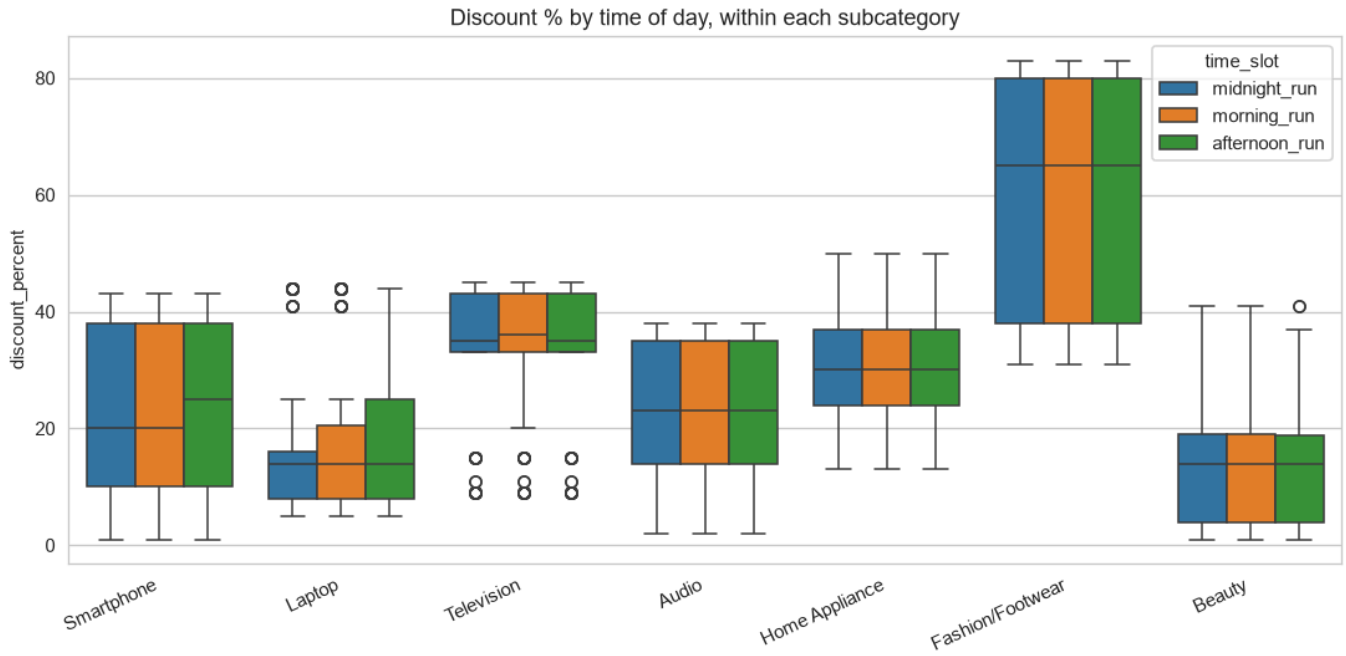
Within each subcategory, how does each festival window compare to others?


Avg discount %: Subcategory × Festival
Audio
Beauty
Fashion/Footwear
Home Appliance
Laptop
Smartphone
Television
subcategory
24.1
22.1
23.5
21.3
16.7
14.2
15.4
15.8
58.0
60.0
58.9
61.0
31.8
29.8
29.6
28.4
17.6
17.4
17.5
18.3
24.8
22.6
23.4
22.3
32.0
33.6
33.8
33.3
Ganesh Chaturthi
Janmashtami
None
Onam + Raksha Bandhan
festival_label
60
50
40
30
20


Avg price change %: Subcategory × Festival
Audio
Beauty
Fashion/Footwear
Home Appliance
Laptop
Smartphone
Television
subcategory
-0.12
0.01
0.05
0.00
0.22
0.00
-0.02
0.00
-0.02
0.46
0.01
0.21
0.00
0.00
0.01
-0.22
0.26
0.14
0.01
0.00
-0.02
-0.01
0.08
0.07
0.08
-0.15
0.06
0.03
Ganesh Chaturthi
Janmashtami
None
Onam + Raksha Bandhan
festival_label
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2

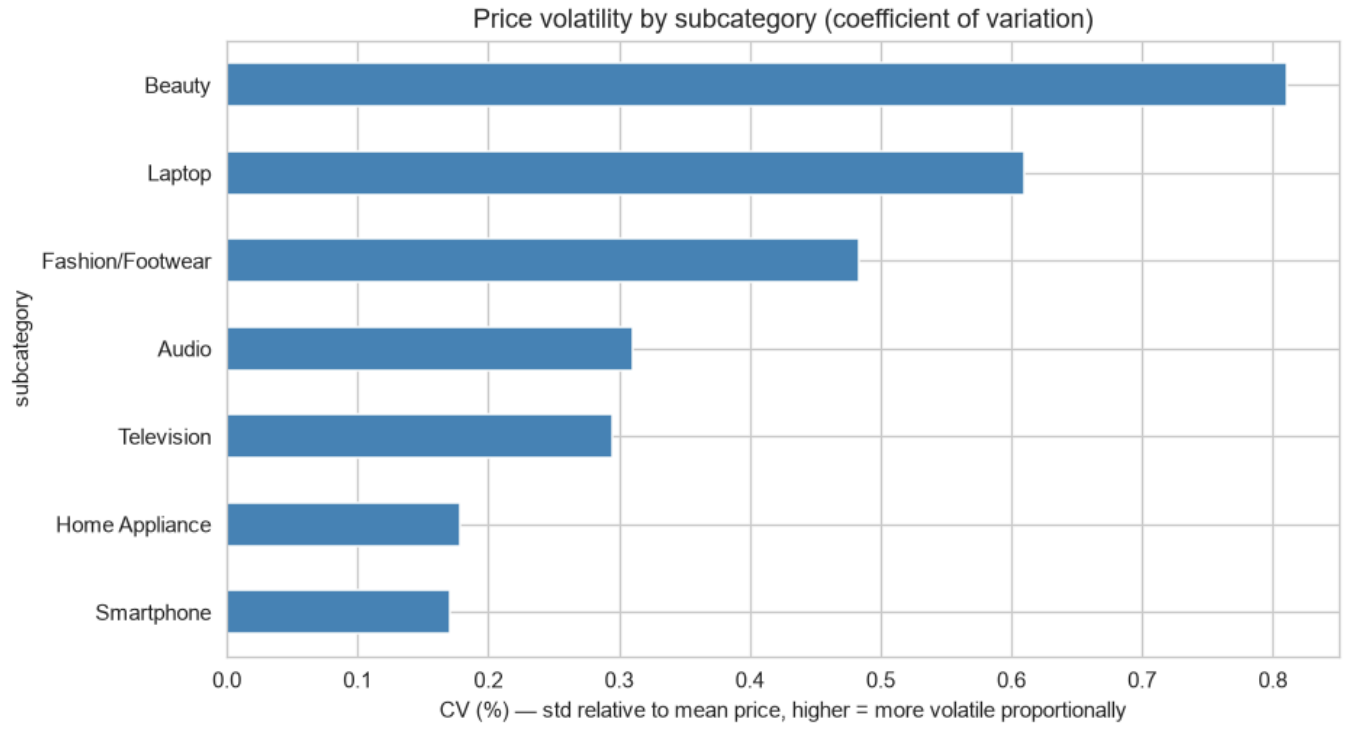
5. Time of Day

Within each subcategory, does the scrape time (12am/8am/4pm) show any difference?



6. Price Volatility

Which subcategory fluctuates most, PROPORTIONALLY to its own price?



Using CV (std ÷ mean × 100) instead of raw std, so expensive products don't automatically rank as "most volatile" just because their ₹ swings are bigger in absolute terms.

---

7. Individual Product Spotlight

Which SPECIFIC products fluctuate most/least, proportionally?

Most volatile (by CV)

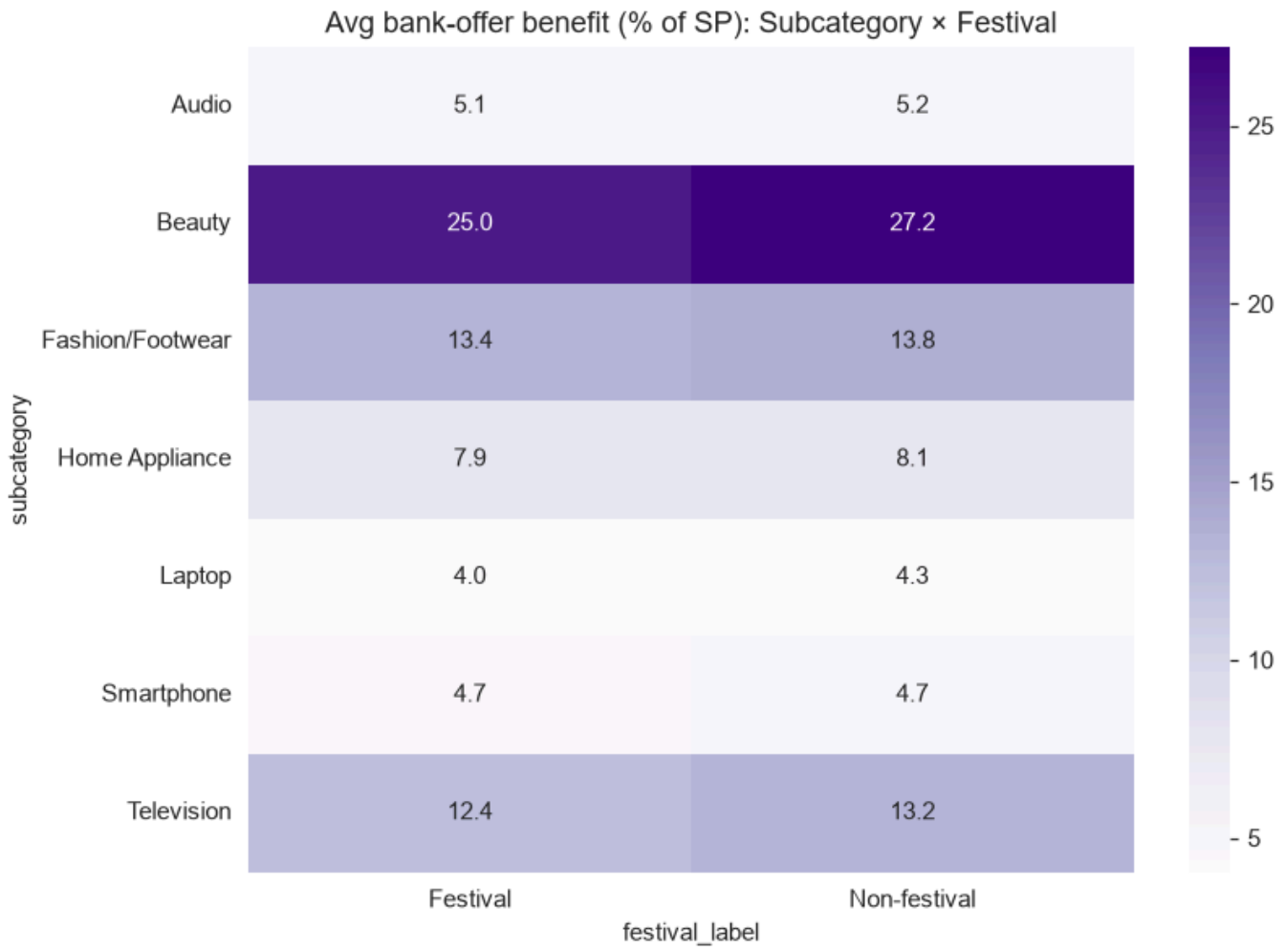
product_id	product_name	subcategory	avg_cv_pct
32	Bata Derby For Men (Brown , 8)	Fashion/Footwear	3.53
11	ASUS ROG Strix G16 (2025) with Office 2024 + M365 Basic* Intel Core i7 14th Gen 14650HX - (16 GB/1 TB SSD/Windows 11 Home/8 GB Graphics/NVIDIA GeForce RTX 5050/240 Hz) G615JHR-S5005WS Gaming Laptop	Laptop	2.36
40	DOVE Nourishing Triple Hydration Serum Body Wash	Beauty	2.13
31	PUMA X-Ray 2 Squa Sneakers For Women	Fashion/Footwear	1.46
7	realme P4 Lite 5G (Mosaic Green, 128 GB) (6 GB RAM)	Smartphone	1.04

Least volatile (by CV)

product_id	product_name	subcategory	avg_cv_pct
33	RED TAPE Men Slides (Grey , 11)	Fashion/Footwear	0.0
38	Lakmé Sunscreen - SPF 50 PA+++ Sun Expert Tinted Sunscreen| UVA/B	Beauty	0.0
37	Garnier Men Oil Clear Deep Cleansing, with Mineral Clay and Menth	Beauty	0.0
39	NIVEA Cocoa Nourish, Moisturizer Body Lotion, Coconut Oil, Serum,	Beauty	0.0
40	DOVE Nourishing Triple Hydration Serum Body Wash (300 ml)	Beauty	0.0

8. Bank Offer Analysis

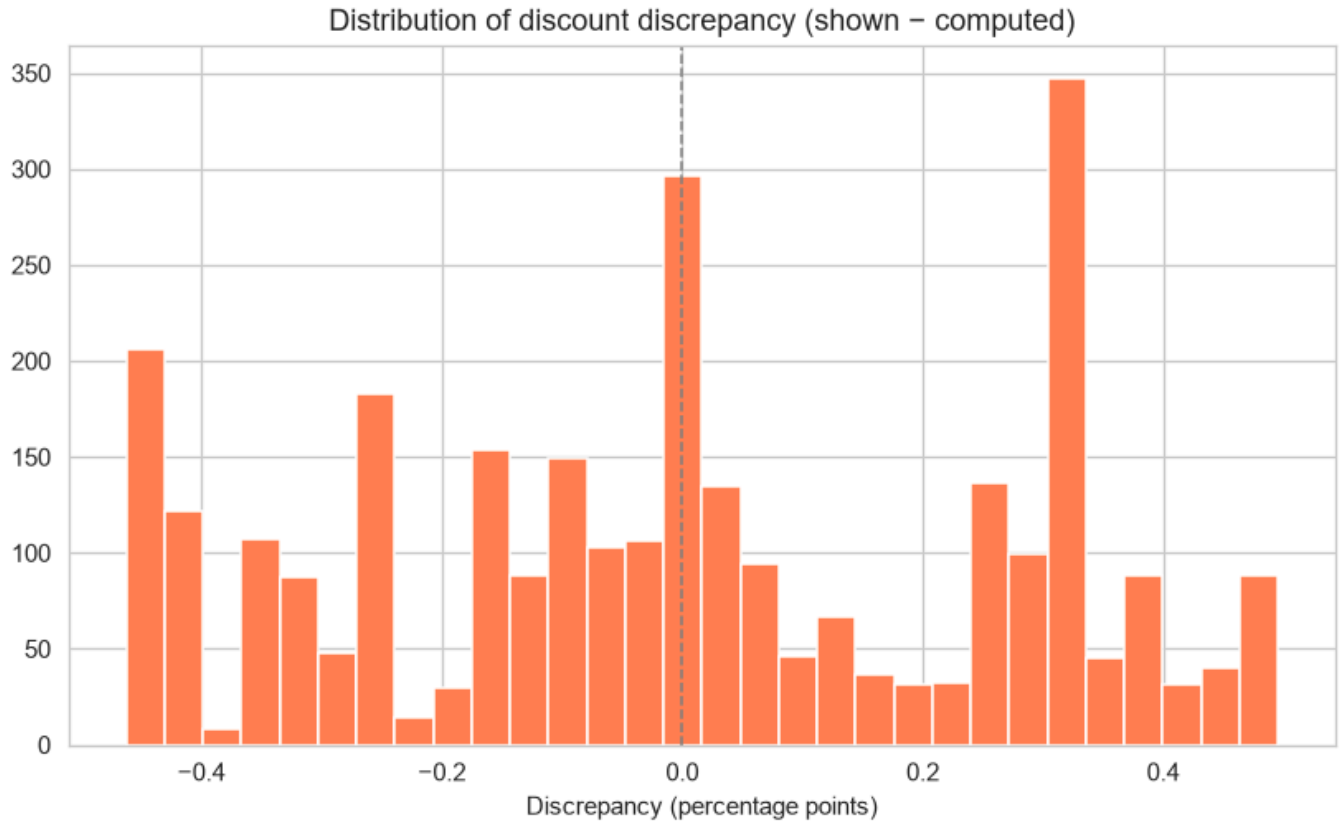
Is the bank-offer benefit ALSO bigger during festivals, and does it vary by subcategory?

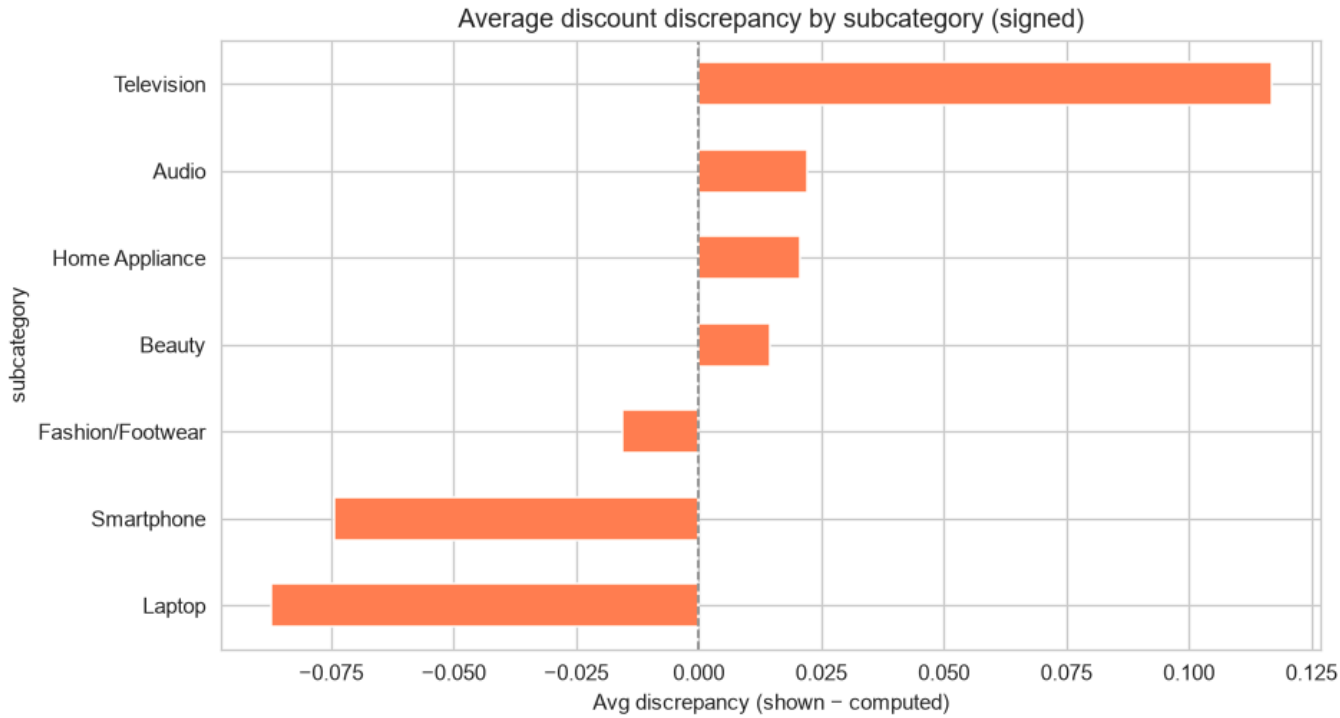


Based on 3410 rows where a bank offer was actually shown (excludes 53 rows with no bank offer, rather than treating missing as ₹0). Shown as % of SP so subcategories at different price levels are comparable.

9. Discount Discrepancy — Overall + By Subcategory

How often, and by how much, does the shown discount disagree with the real one?

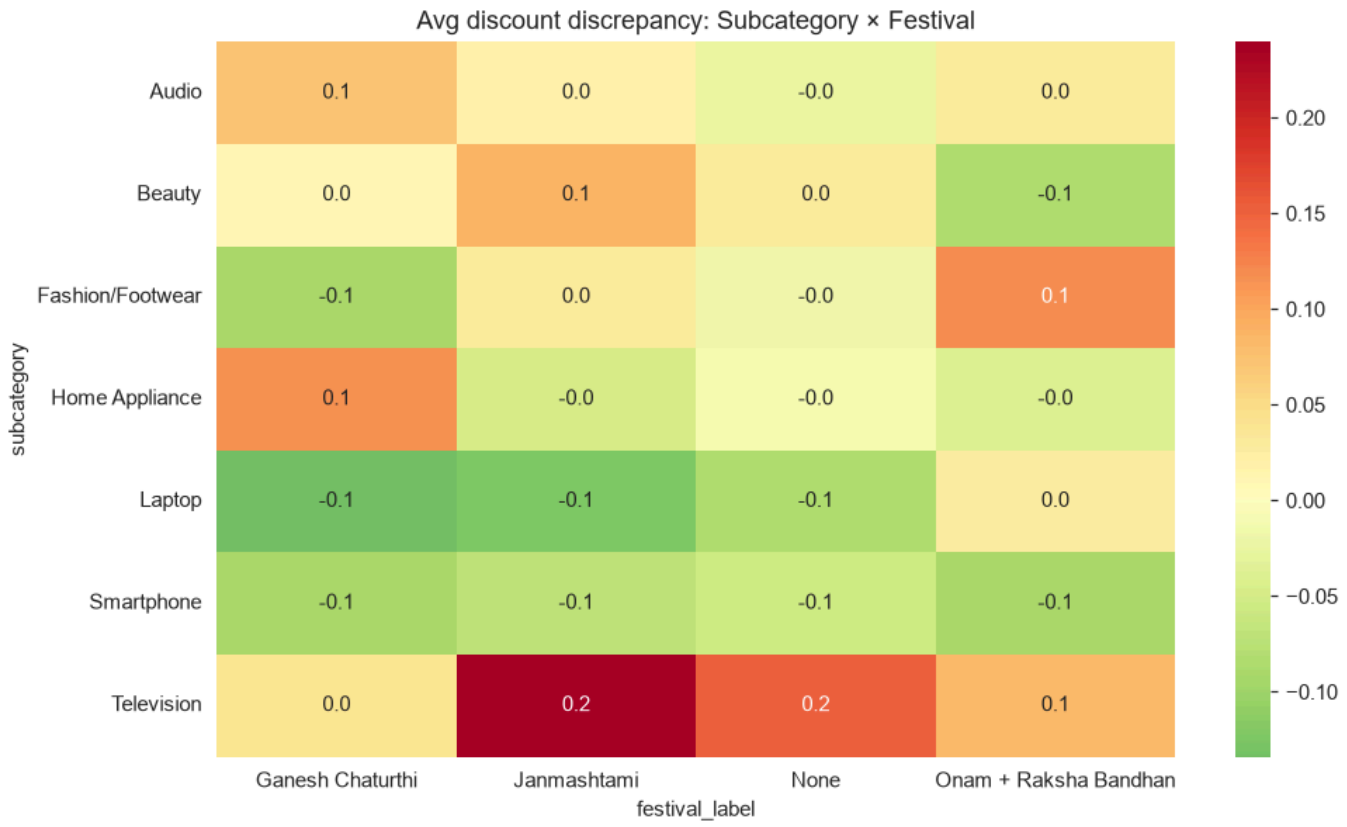


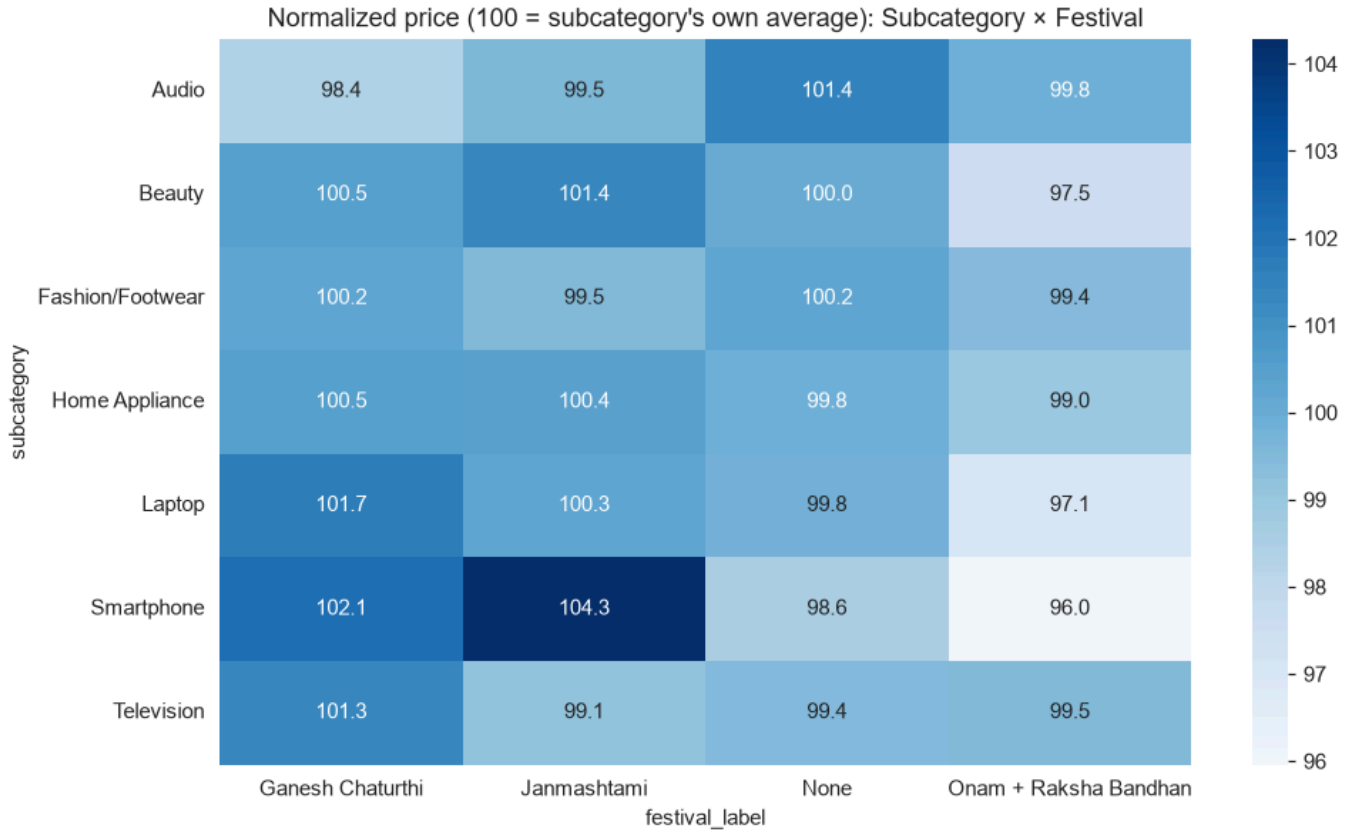


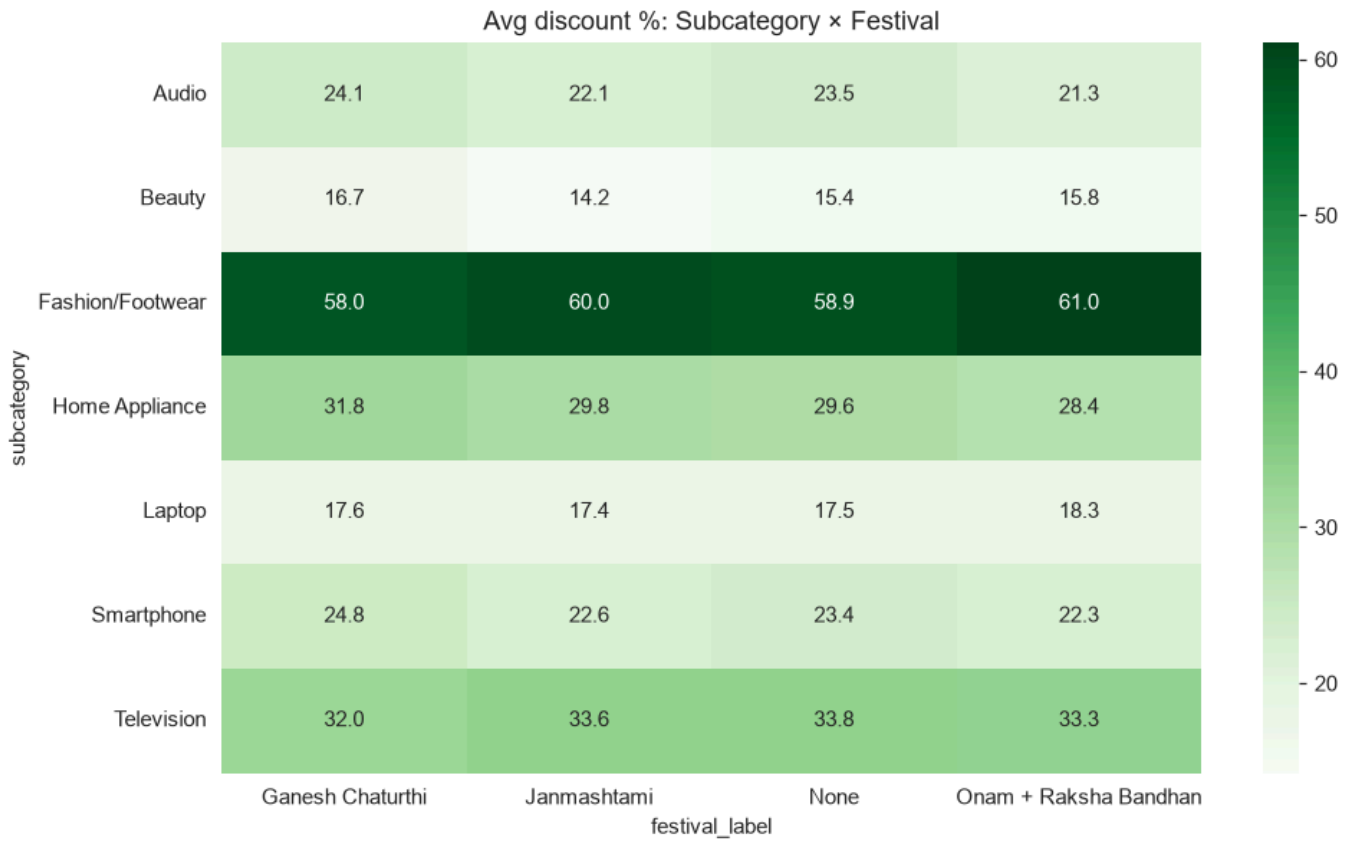
---

10. Subcategory × Festival Cross-Check

Where does discount discrepancy concentrate — and does it worsen during festivals?


Avg discount discrepancy: Subcategory × Festival
Audio
Beauty
Fashion/Footwear
Home Appliance
Laptop
Smartphone
Television
subcategory
0.1 0.0 -0.0 0.0
0.0 0.1 0.0 -0.1
-0.1 0.0 -0.0 0.1
0.1 -0.0 -0.0 -0.0
-0.1 -0.1 -0.1 0.0
-0.1 -0.1 -0.1 -0.1
0.0 0.2 0.2 0.1
Ganesh Chaturthi
Janmashtami
None
Onam + Raksha Bandhan
festival_label
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10


Normalized price (100 = subcategory's own average): Subcategory × Festival
Audio
Beauty
Fashion/Footwear
Home Appliance
Laptop
Smartphone
Television
subcategory
98.4 99.5 101.4 99.8
100.5 101.4 100.0 97.5
100.2 99.5 100.2 99.4
100.5 100.4 99.8 99.0
101.7 100.3 99.8 97.1
102.1 104.3 98.6 96.0
101.3 99.1 99.4 99.5
Ganesh Chaturthi
Janmashtami
None
Onam + Raksha Bandhan
festival_label
104
103
102
101
100
99
98
97
96

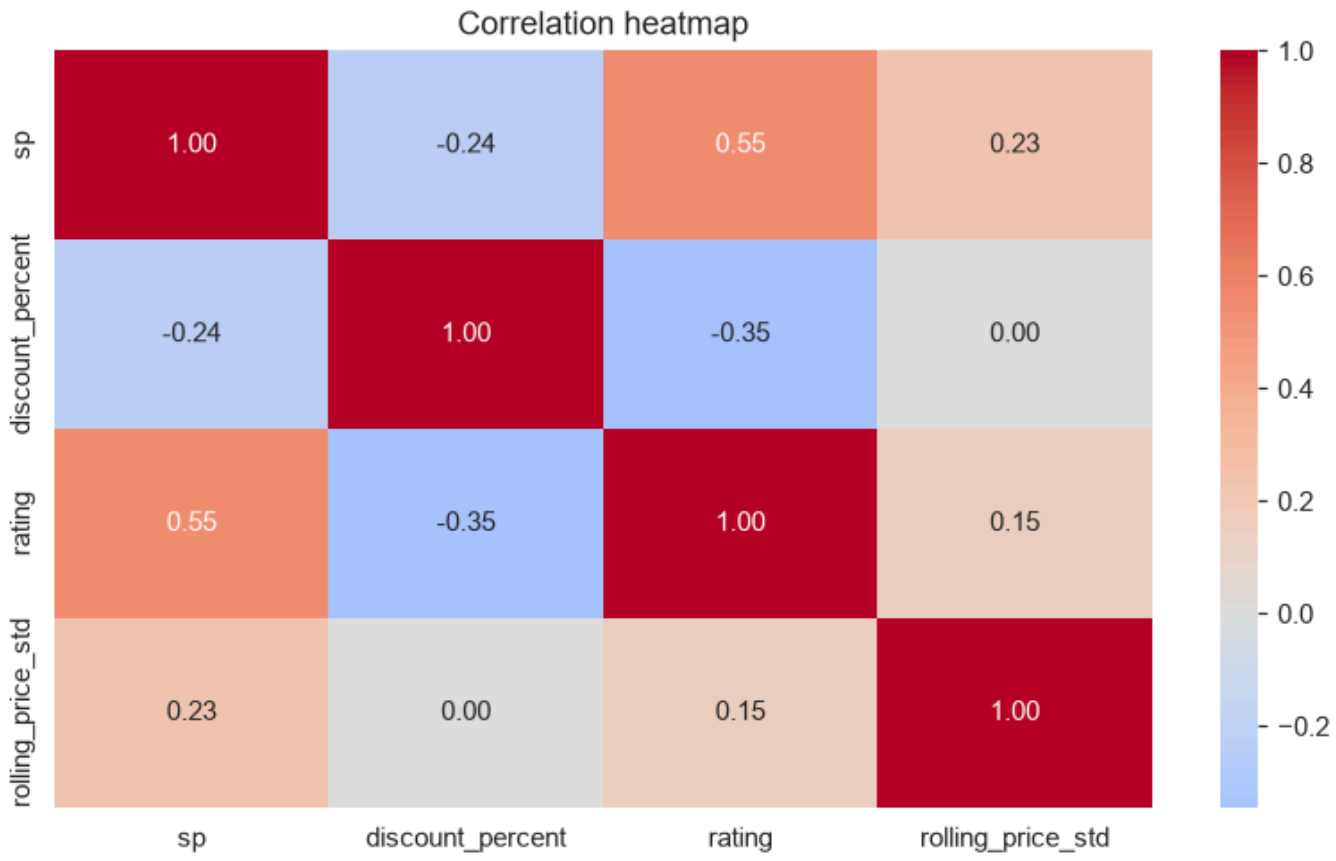


These three views are best read together, not the discrepancy heatmap alone: a discrepancy value on its own doesn't say whether it reflects genuine discounting or an inflated MRP claim — the normalized price and discount% patterns alongside it help narrow down which is more likely. Any interpretation here is suggestive, not conclusive.

---

11. Correlation Heatmap

Do price, discount, rating, and volatility relate to each other?


Correlation heatmap
sp
discount_percent
rating
rolling_price_std
1.00
-0.24
0.55
0.23
-0.24
1.00
-0.35
0.00
0.55
-0.35
1.00
0.15
0.23
0.00
0.15
1.00
sp
discount_percent
rating
rolling_price_std
1.0
0.8
0.6
0.4
0.2
0.0
−0.2